Phosphorus diffusion into silicon

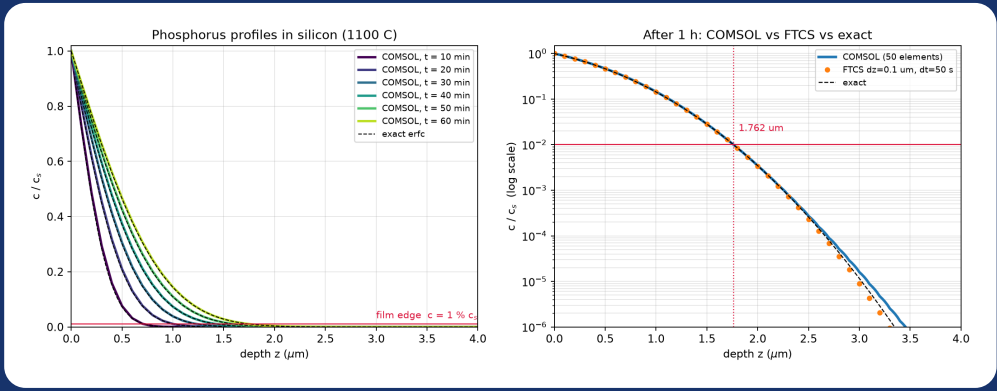
Step-by-step COMSOL Multiphysics 4.3a guide for students

Problem: 1D unsteady diffusion in a semi-infinite medium (Introduction to CFD, Problem 2)

Tool: COMSOL Multiphysics 4.3a, Transport of Diluted Species, 1D, time-dependent study

Result: film depth after 1 h = 1.762 um (exact 1.762 um); COMSOL curves match the erf solution

Includes: 14 annotated steps, theory, FTCS stability analysis and verification



Model file: phosphorus_diffusion.mph (open it in COMSOL to compare with your own model)

The problem and the solution plan

In a CVD reactor the P-rich SiO2 coating on a silicon wafer supplies phosphorus. The phosphorus diffuses into the wafer, which starts free of P. The coating keeps the surface concentration constant, and the wafer is semi-infinite. Predict the depth of the Si-P film after 1 h (film edge: 1 % of the surface value) and the concentration profile after 1 h.

Quantity	Symbol	Value
process temperature	T	1100 C
surface concentration	c_As	2.5e20 atoms/cm^3 (415.13 mol/m^3)
initial concentration	c_A0	0
diffusivity of P in Si	D_AB	6.5e-13 cm^2/s = 6.5e-17 m^2/s
doping time	t	1 h = 3600 s
film edge (1 % of surface)	c_A	2.5e18 atoms/cm^3 (4.1513 mol/m^3)

Assumptions: unsteady state; semi-infinite medium; 1D diffusion in z; constant boundary conditions; constant temperature, pressure and diffusivity.

How this guide solves it

1. Exact solution (error function): film depth by hand.
2. COMSOL: Transport of Diluted Species, time-dependent, 1D.
3. Check COMSOL against the exact solution.
4. Finite-difference FTCS check and stability limit (Python).

1. Physics: mass balance and Fick's law

$$\frac{\partial c_A}{\partial t} = D_{AB}\frac{\partial^2 c_A}{\partial z^2}$$

Initial condition: c_A(z, 0) = c_A0 = 0.

Boundary conditions: c_A(0, t) = c_As (source), c_A(z -> infinity, t) = 0 (sink).

2. Exact solution (similarity variable z / 2 sqrt(D t))

$$\frac{c_{As} - c_A}{c_{As} - c_{A0}} = \operatorname{erf}\left(\frac{z}{2\sqrt{D_{AB}\,t}}\right)$$

$$z_{film} = 2\sqrt{D_{AB}\,t}\,\operatorname{erfc}^{-1}(0.01) = 1.8214 \times 2\sqrt{D_{AB}\,t}$$

With D = 6.5e-17 m^2/s and t = 3600 s: sqrt(D t) = 0.4837 um, so z_film = 1.8214 x 0.9675 um = 1.7621 um.

3. Domain and numerical choices

z_max = 5 um: there c/c_s = erfc(5.17) ~ 4e-13 after 1 h, so the cut-off hardly matters.

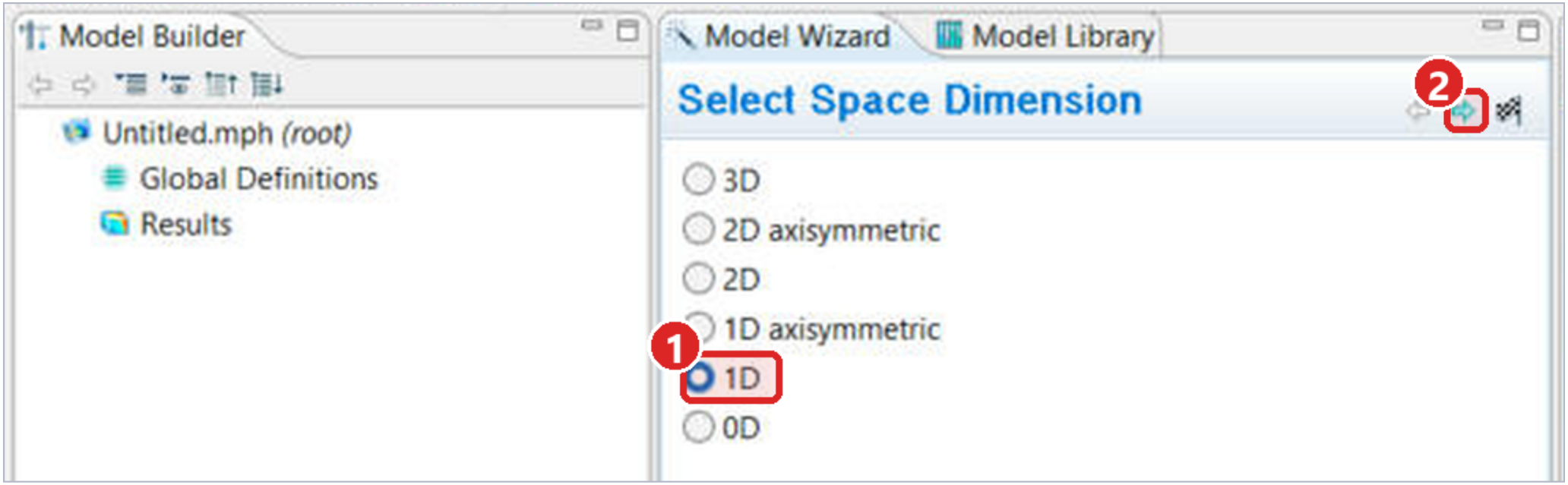
dz = 0.1 um (50 intervals) as asked in the task.

COMSOL: implicit time stepper (no stability limit).

FTCS: dt <= dz^2/(2 D) = 76.9 s, we use dt = 50 s (Fo = D dt/dz^2 = 0.325).

Step 1 - Start a 1D model

Start COMSOL Multiphysics 4.3a. The Model Wizard opens (if not: File > New).

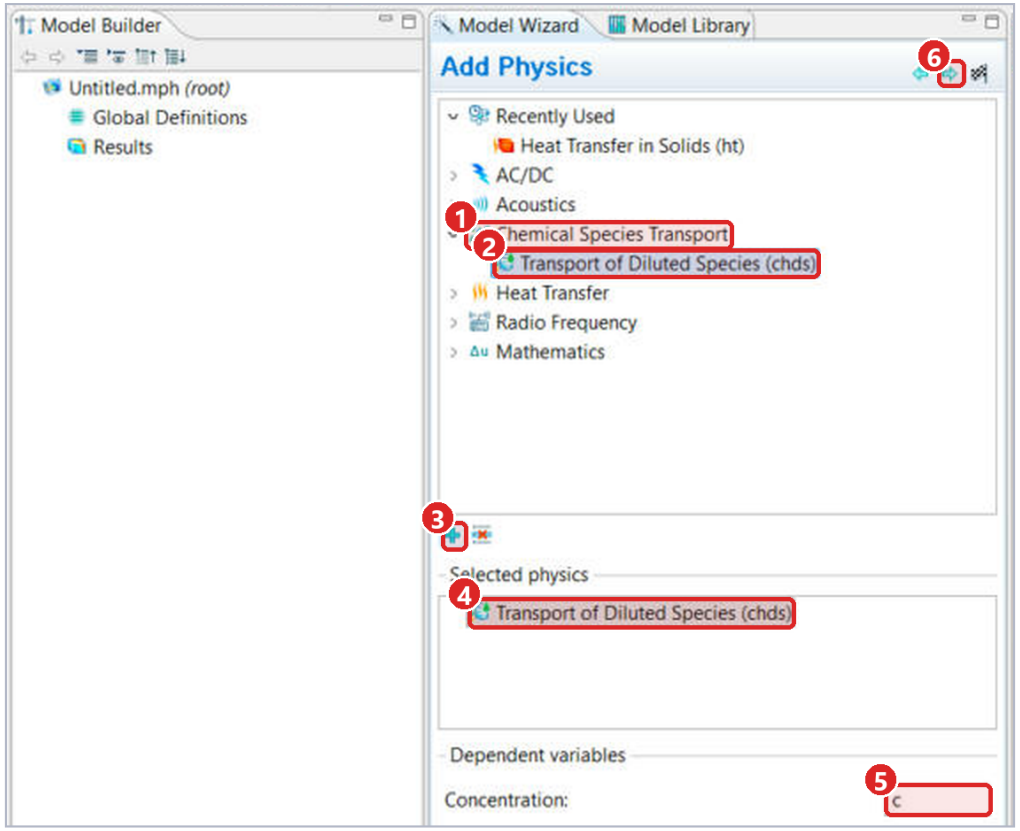


1. Select 1D. Diffusion into the wafer happens only along the depth direction z (Assumption 3). COMSOL calls the 1D coordinate x, so x in COMSOL means the depth z of the problem.
2. Click the Next arrow.

Why?
A 1D model needs only a line: the surface of the wafer is one end point and the deep silicon the other end point.

Step 2 - Add Transport of Diluted Species

Diffusion of phosphorus (a dilute species) in silicon is mass transport by Fick's law.

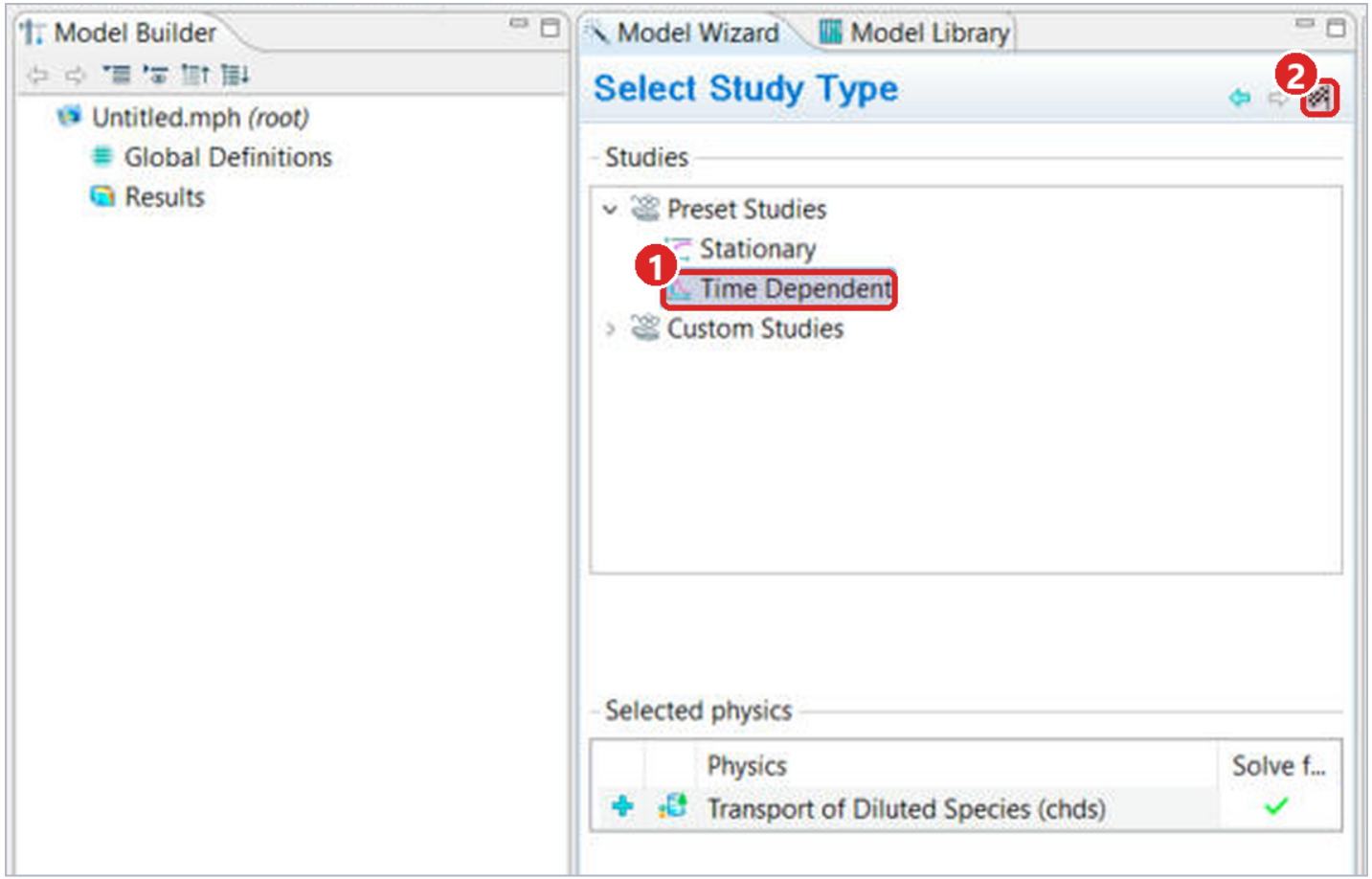


1. Expand Chemical Species Transport (double-click it).
2. Select Transport of Diluted Species (chds).
3. Click the blue + (Add Selected) button.
4. The interface appears under Selected physics.
5. The dependent variable is the concentration c (unit mol/m^3).
6. Click Next.

Why?
The interface solves dc/dt + div(-D grad c + u c) = R. With zero velocity (u = 0) and no reaction (R = 0) it reduces to the diffusion equation dc/dt = D d2c/dz2 of the problem.

Step 3 - Choose a time-dependent study

The concentration profile changes with time, so we need a transient study.

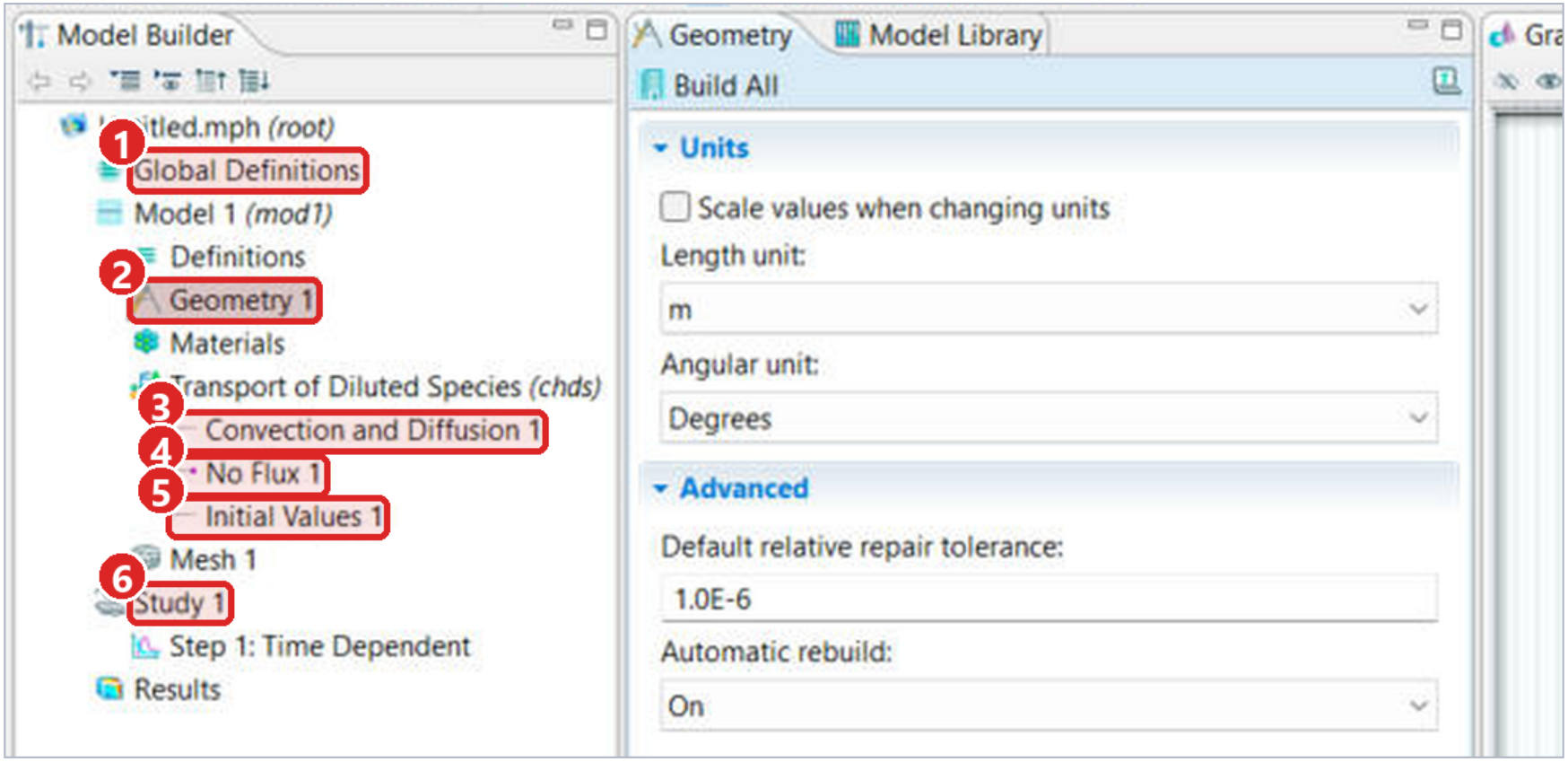


1. Select Time Dependent (under Preset Studies).
2. Click Finish (the chequered-flag icon).

Why?
Unsteady state is Assumption 1 of the problem. COMSOL will march the solution from t = 0 to t = 1 h with an automatic implicit time stepper. This replaces the explicit FTCS scheme of the finite-difference approach.

Step 4 - Tour of the model tree

After Finish the Model Builder holds the model outline. Note what COMSOL has already created for us.

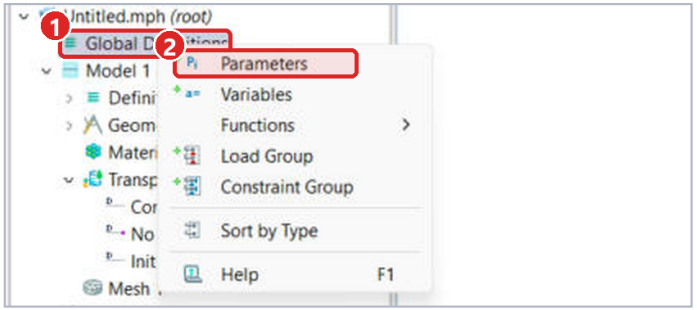


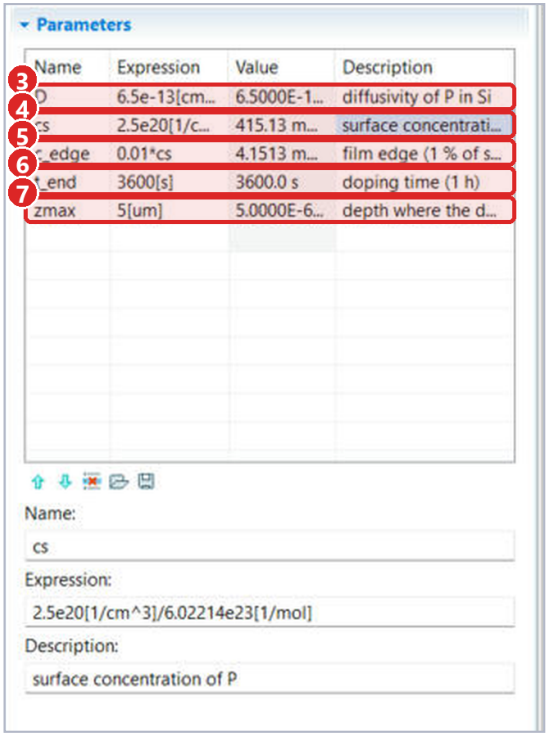
1. Global Definitions: we will store the problem data here as parameters.
2. Geometry 1: the line representing the wafer depth.
3. Convection and Diffusion 1: the equation. We only enter the diffusivity D.
4. No Flux 1 is the default on every boundary. We will override it with Concentration conditions.
5. Initial Values 1: starting concentration (we check that it is zero).
6. Study 1: the time-dependent study.

Why?
A material is not needed here, because the diffusivity is entered directly. The length unit is metre, so the depth values will appear in metres.

Step 5 - Enter the problem data as parameters

Parameters keep all numbers in one place, so students can change them later and rerun.



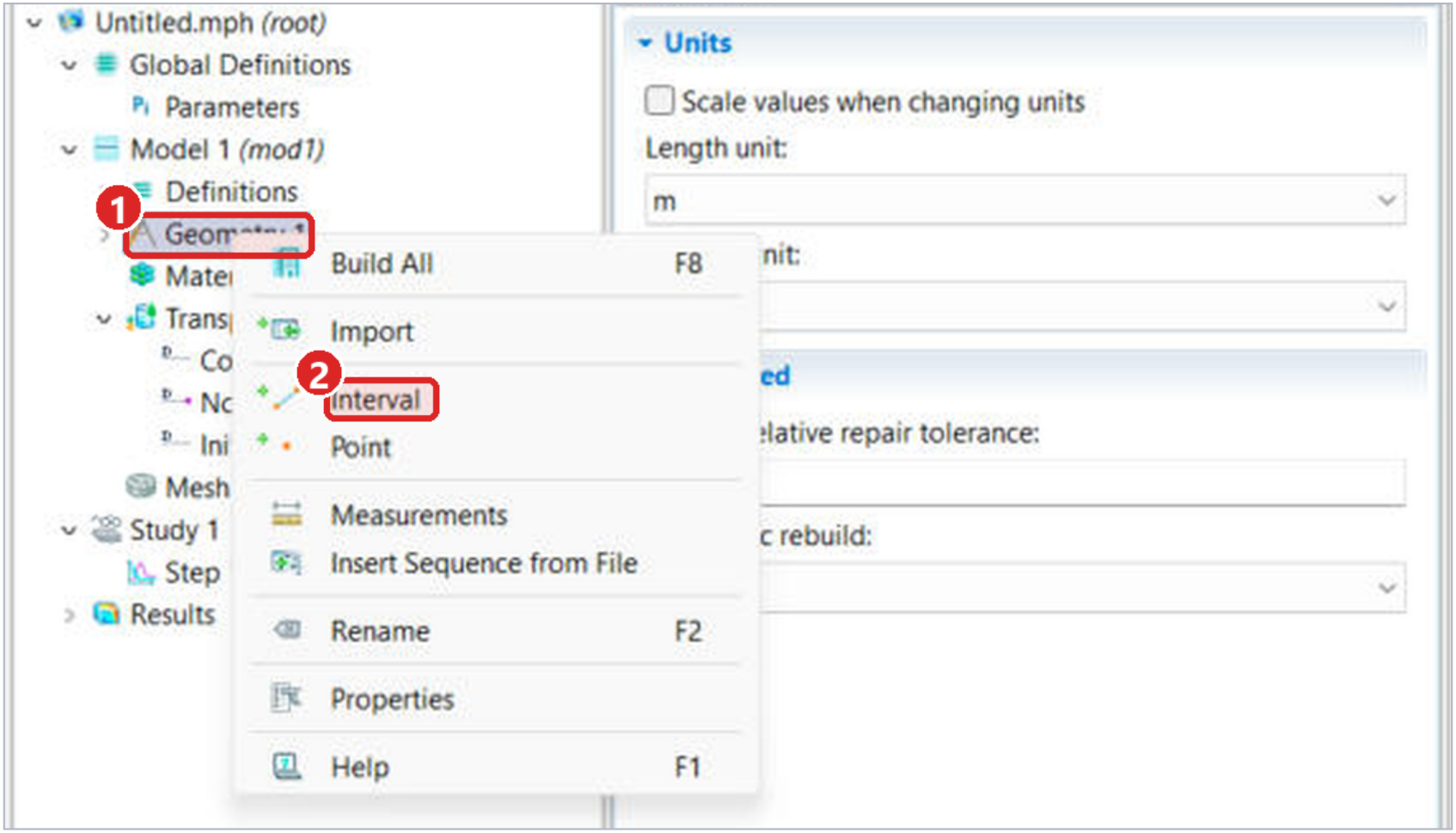


1. Right-click Global Definitions (select it first if the menu does not appear).
2. Choose Parameters.
3. D = 6.5e-13[cm^2/s]: diffusivity of P in Si at 1100 C. COMSOL converts it to 6.5e-17 m^2/s.
4. cs = 2.5e20[1/cm^3]/6.02214e23[1/mol]: surface concentration 2.5e20 atoms/cm^3 expressed in mol/m^3 (= 415.13 mol/m^3). COMSOL's species concentration is in mol/m^3, so we divide by Avogadro's number.
5. c_edge = 0.01*cs: the film edge, 1 % of the surface value (4.1513 mol/m^3).
6. t_end = 3600[s]: doping time of 1 h.
7. zmax = 5[um]: depth at which we cut the semi-infinite wafer. Type each Name, press Tab, type the Expression, press Tab again for the Description.

Why?
Units are written in square brackets. The diffusion length after 1 h is sqrt(D t) = 0.48 um and the profile reaches about 2 sqrt(D t) = 0.97 um before it has dropped by a factor 2.7. At z = 5 um the exact profile is c/cs = erfc(5.17), about 4e-13, i.e. zero for practical purposes: so 5 um is deep enough to represent the semi-infinite wafer.

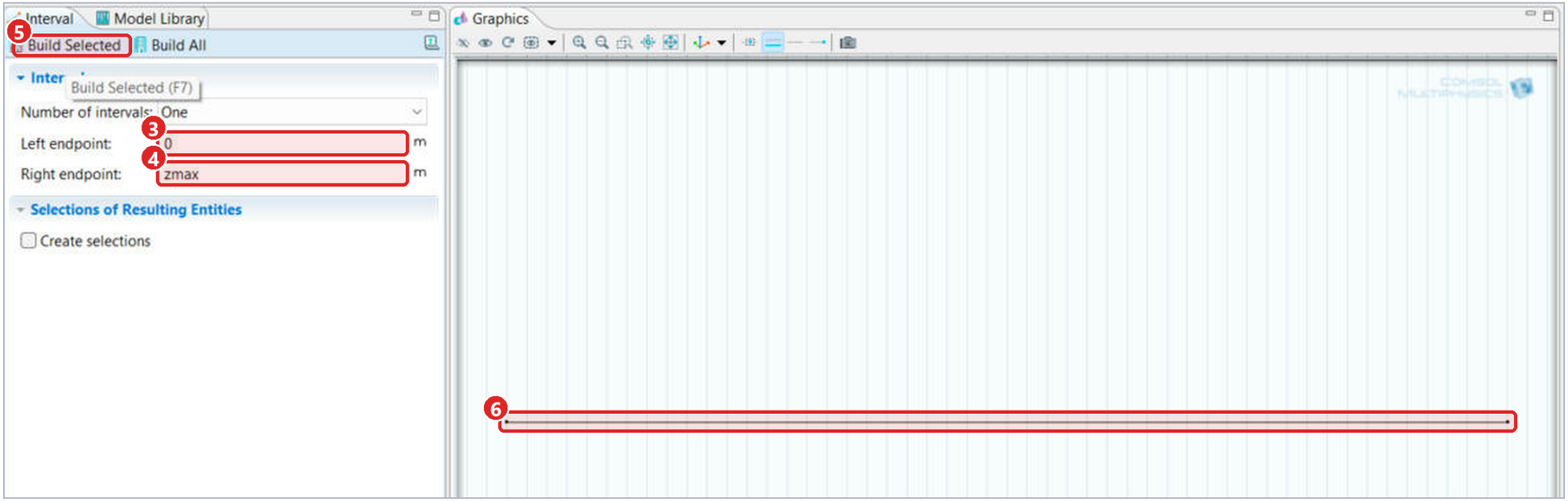
Step 6 - Build the geometry: a 5 um line

The wafer is semi-infinite, but a computer model needs a finite depth. We use z from 0 (surface) to zmax = 5 um.



1. Right-click Geometry 1.
2. Choose Interval.

Step 6 - Build the geometry: a 5 um line

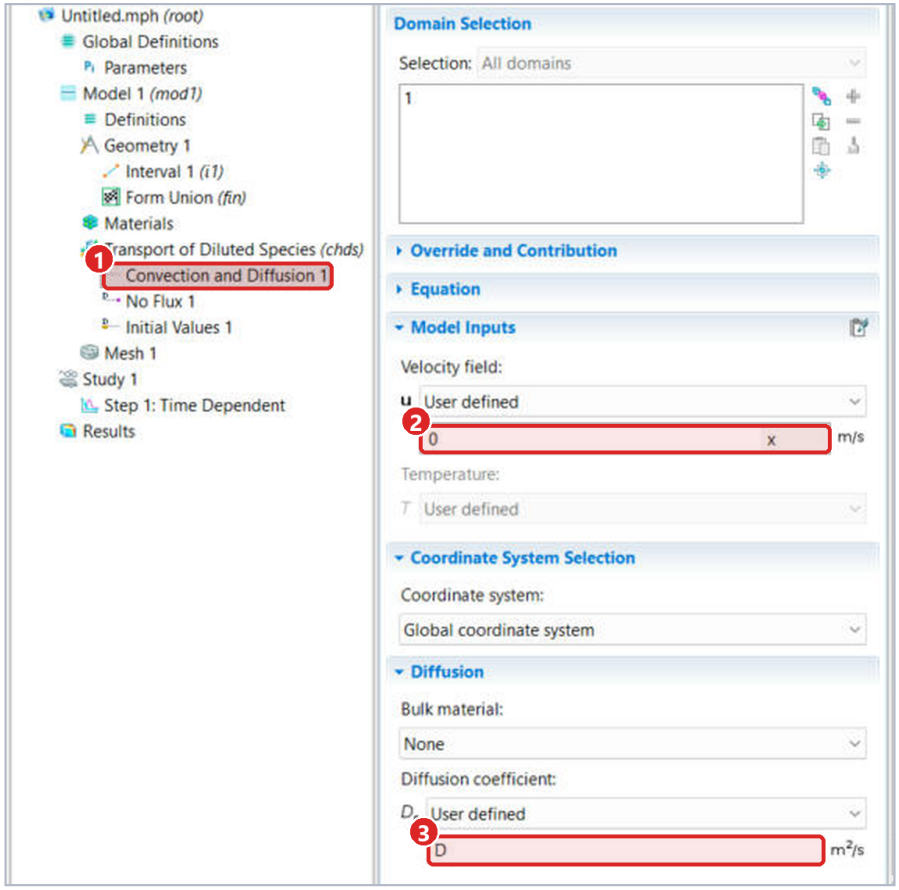


3. Left endpoint = 0 (the wafer surface, z = 0).
4. Right endpoint = zmax (the parameter, 5 um).
5. Click Build Selected (or press F8).
6. The line appears. The axis shows 0 to 50 with the factor x10^-7: that is 0 to 5e-6 m = 5 um. Point 1 is the surface (z = 0), Point 2 is the deep end.

Why?
Why 5 um? A cut-off depth is acceptable when the concentration there is still essentially the initial value (zero). The check above shows c(5 um)/cs of about 4e-13 after 1 h. Using a much deeper domain would only add elements without changing the answer.

Step 7 - Set the diffusion coefficient

We tell COMSOL the value of the diffusivity of phosphorus in silicon.

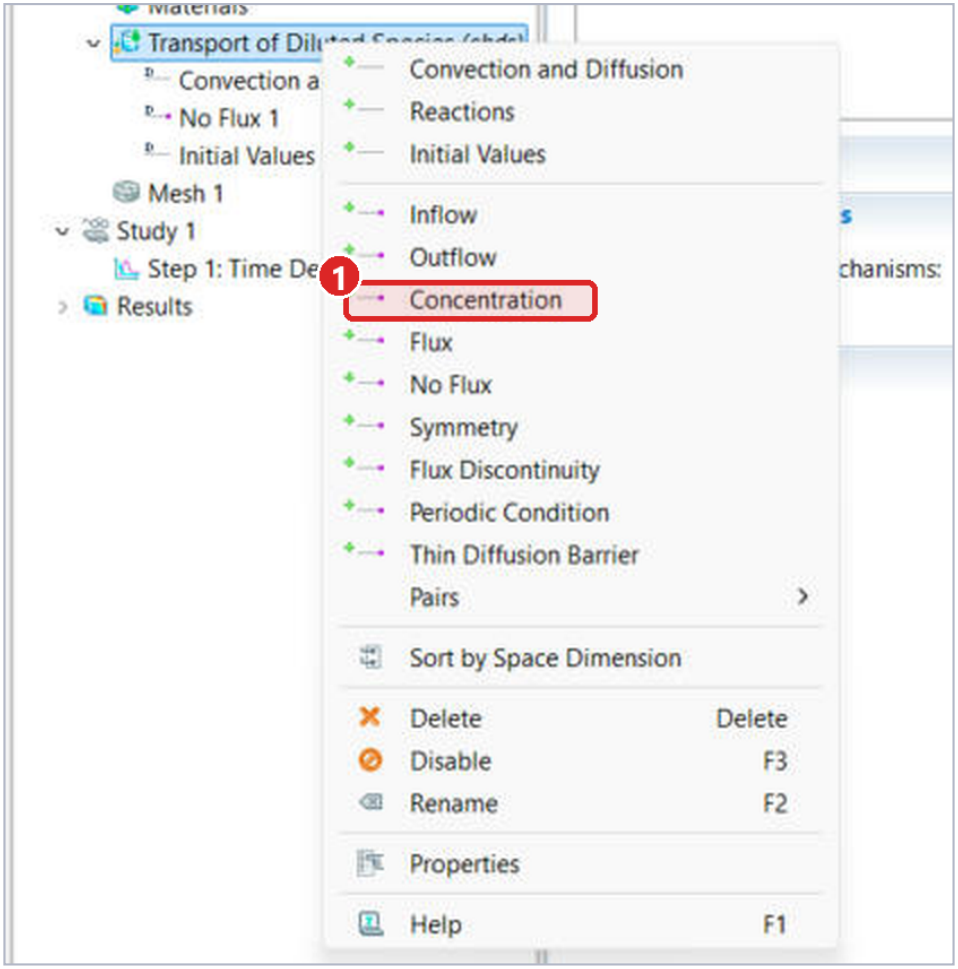


1. Click Convection and Diffusion 1 under Transport of Diluted Species.
2. Velocity field u = 0: there is no convection, only diffusion (the default).
3. Diffusion coefficient D_c: choose User defined and type D (the parameter). The unit m^2/s is shown next to the field.

Why?
The equation solved is dc/dt = d/dz (D dc/dz). D is constant (Assumption 5), so it is the same everywhere in the wafer. The P-rich coating is not modelled: it only acts through the fixed surface concentration.

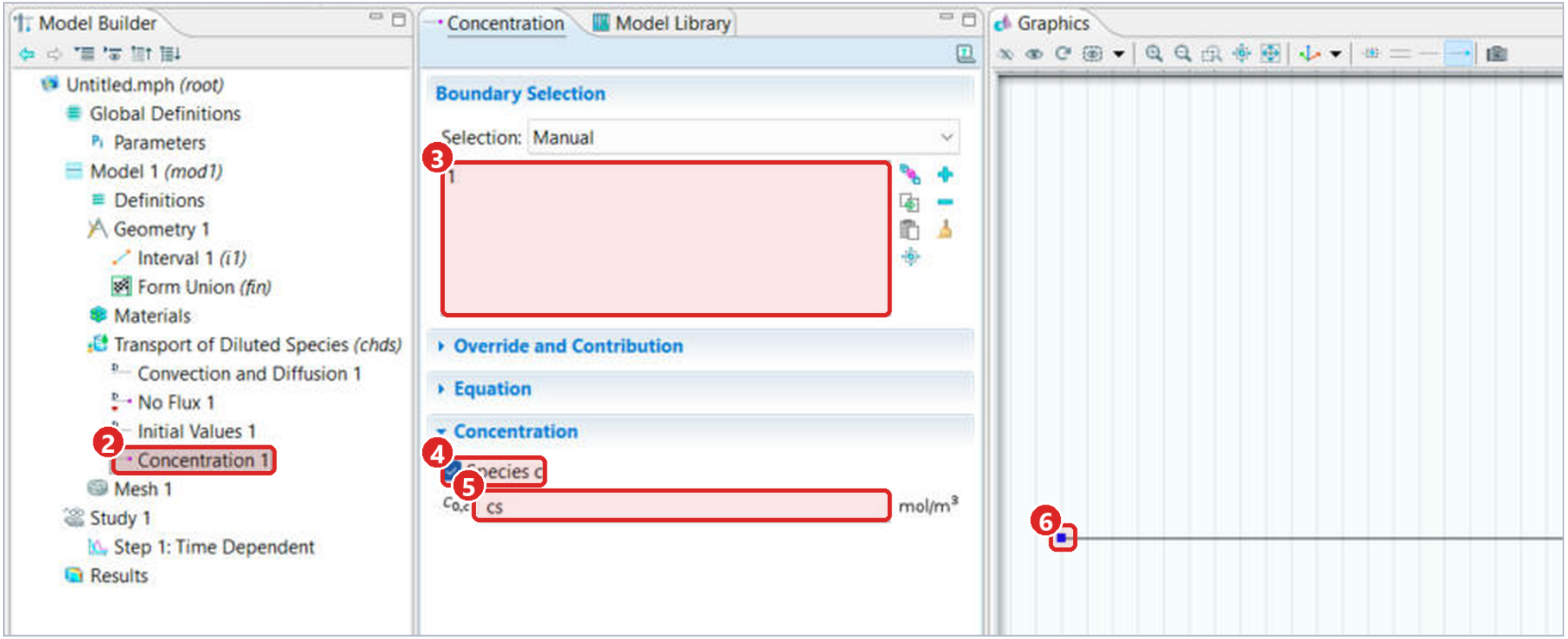
Step 8 - Fix the surface concentration

The P-rich SiO2 coating is an infinite source: the concentration at the silicon surface stays at c_As for all times.



1 Right-click Transport of Diluted Species and choose Concentration.

Step 8 - Fix the surface concentration

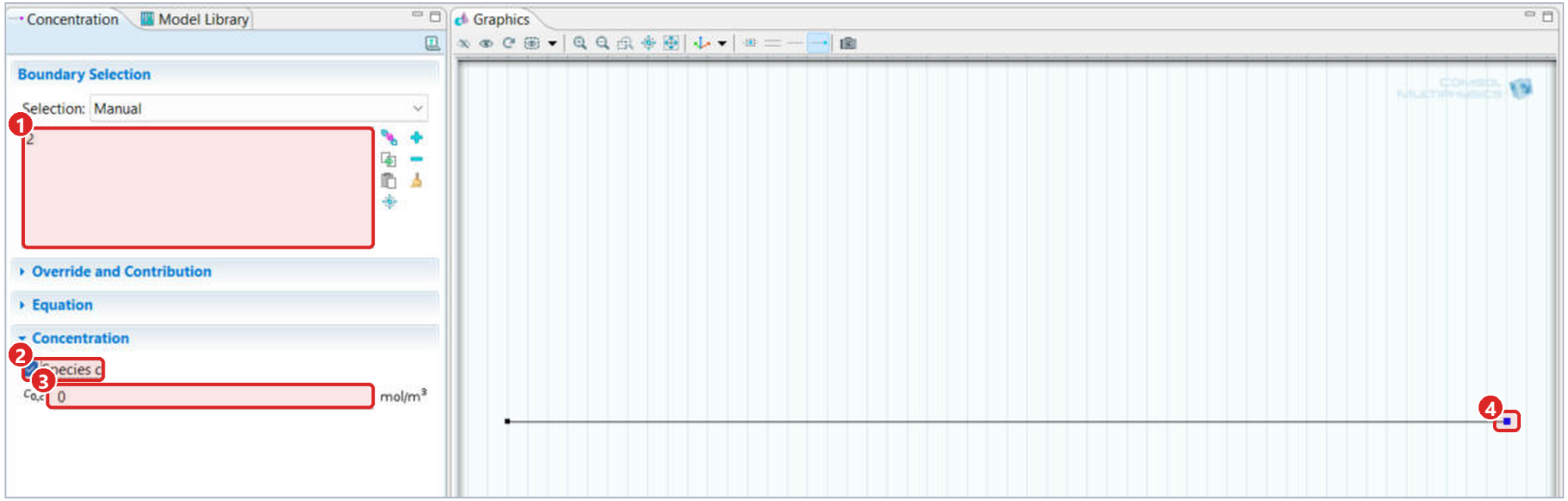


2. The node Concentration 1 appears.
3. The selection list must contain 1 (the surface point). Hover over the left end point until it turns red, click it, then press the + button (Add to Selection).
4. Tick Species c.
5. Concentration c0,c = cs (the parameter, 415.13 mol/m^3).
6. The selected surface point is drawn in blue.

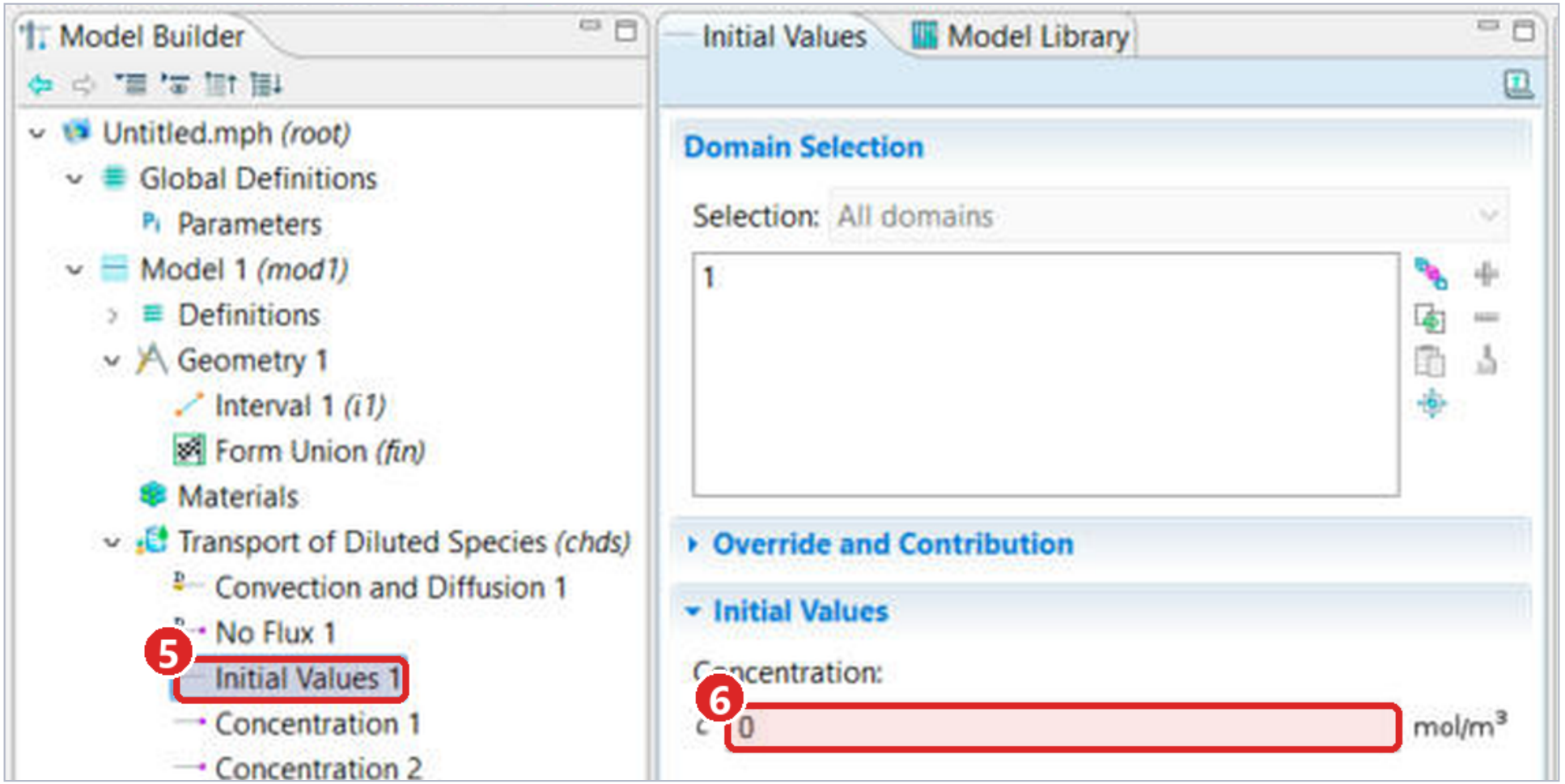
Why?
This is the Dirichlet boundary condition c(z = 0, t) = c_As.

Step 9 - Cut-off boundary and initial condition

At the deep end of the model the concentration is still zero. We also check the initial condition.



1. Add a second Concentration node (right-click Transport of Diluted Species > Concentration). Its selection list must contain 2 (the deep end point).
2. Tick Species c.
3. Keep c0,c = 0: this represents the undisturbed wafer far from the surface, c(zmax, t) = 0 (the semi-infinite sink).
4. The deep end point is shown in blue.



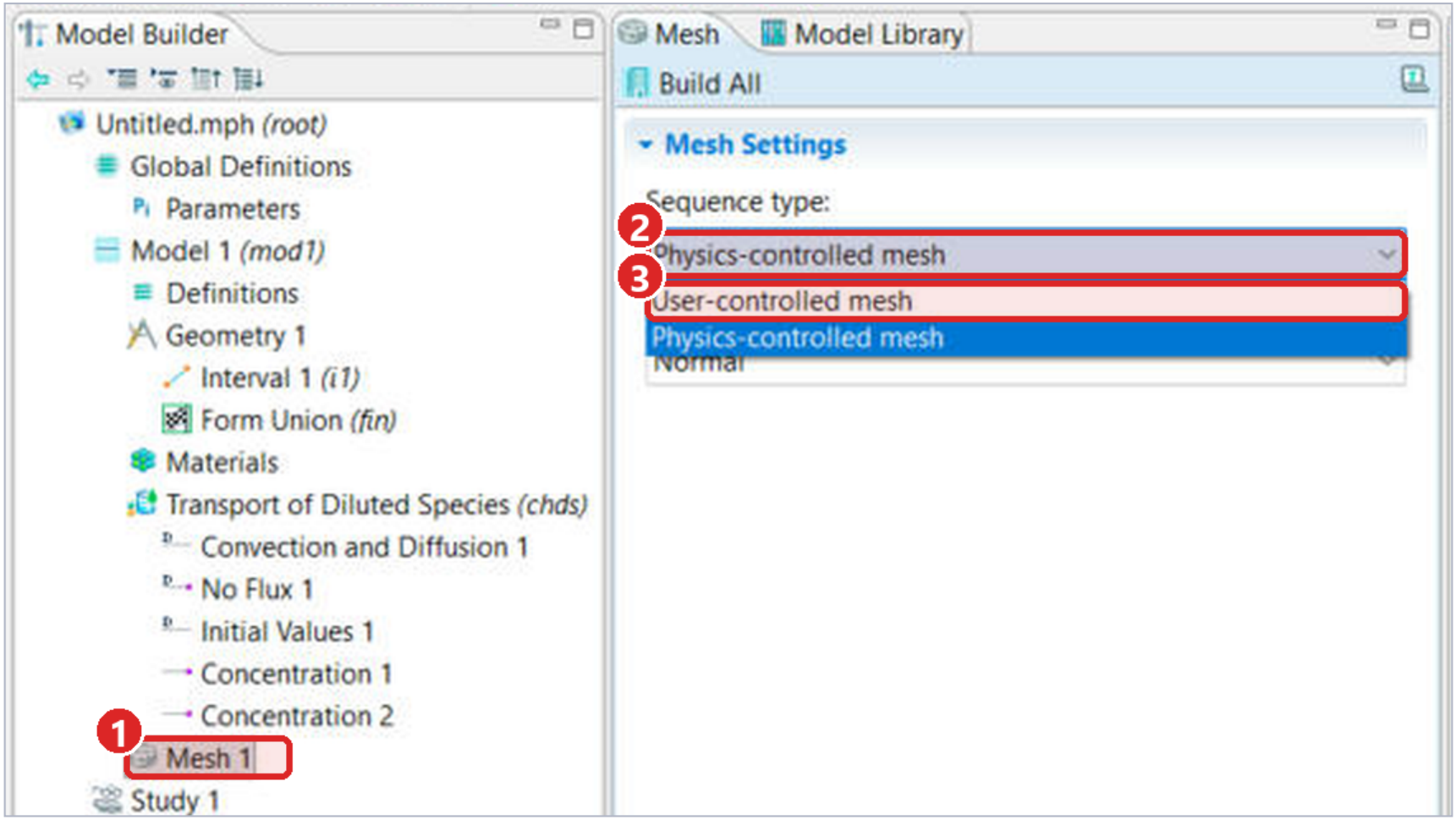
5 Initial Values 1: the wafer starts free of phosphorus.

6 Concentration c = 0 mol/m^3 (default): the initial condition c(z, 0) = c_A0 = 0.

Why?
Together these give the three conditions of the problem: initial c = 0, surface c = c_As, and c -> 0 at depth. All other boundaries keep the default No Flux condition.

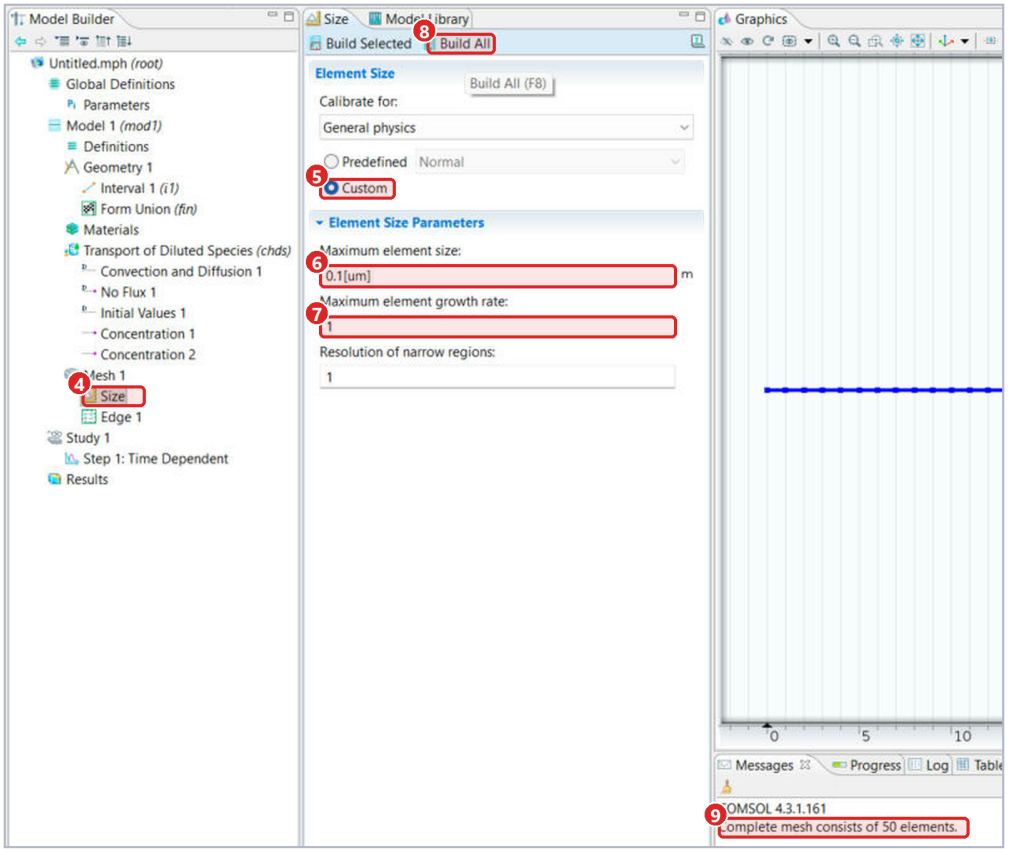
Step 10 - Create the mesh: 0.1 um elements

The task asks for a grid spacing of 0.1 um, so we set the maximum element size to 0.1 um.



1. Click Mesh 1.
2. Open Sequence type.
3. Choose User-controlled mesh. Two nodes appear: Size and Edge 1.

Step 10 - Create the mesh: 0.1 um elements



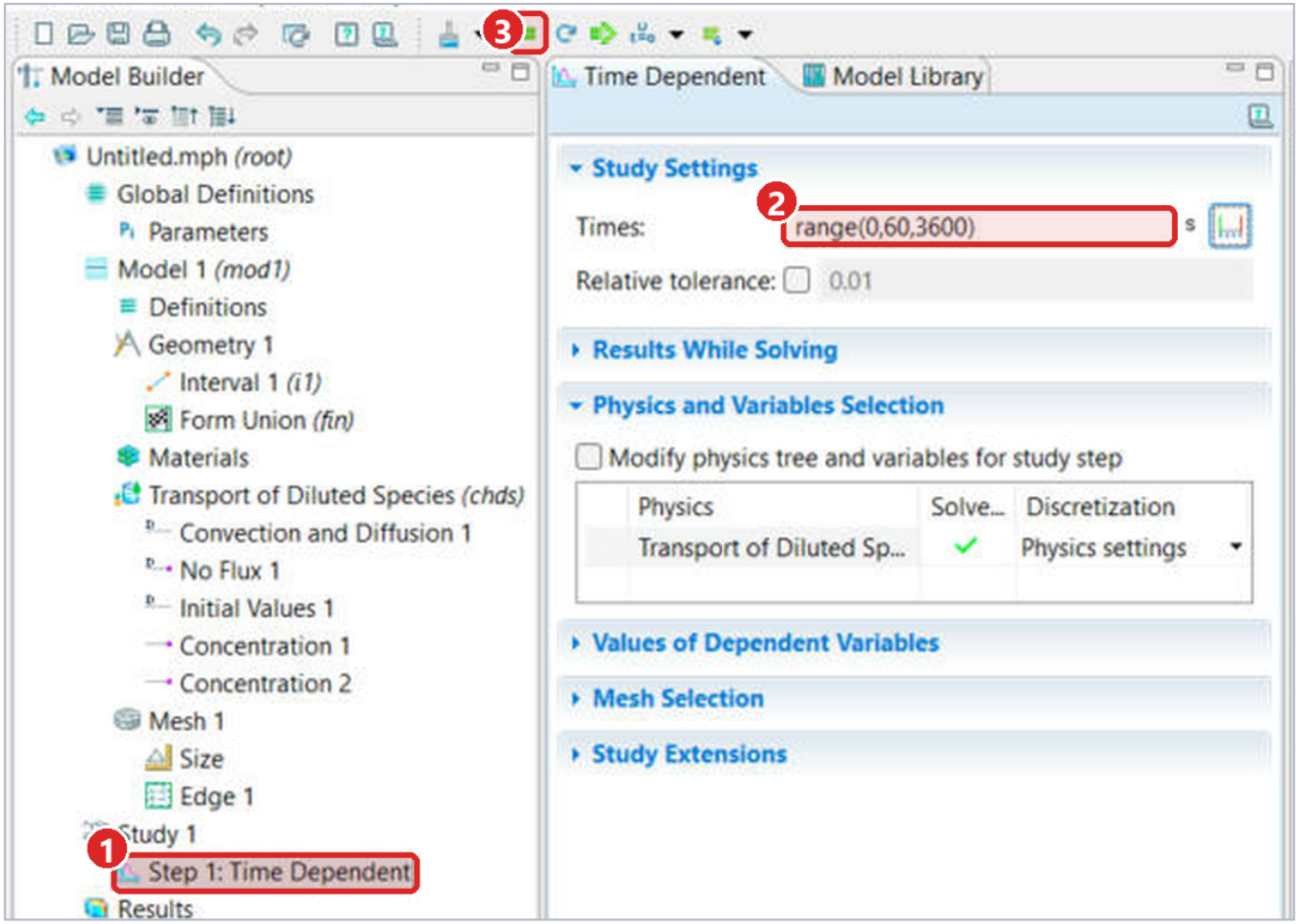
4. Click the Size node.
5. Select Custom.
6. Maximum element size = 0.1[um].
7. Maximum element growth rate = 1 (a uniform mesh).
8. Click Build All (F8).
9. The Messages window reports: Complete mesh consists of 50 elements (5 um / 0.1 um).

Why?

The transport interface uses linear elements by default, so 50 elements give 51 nodes and 51 degrees of freedom (one per node). For a verification study change the maximum element size from 0.1 um to 0.05 um and 0.025 um, compute again and compare the film depth: it should approach the exact value 1.762 um.

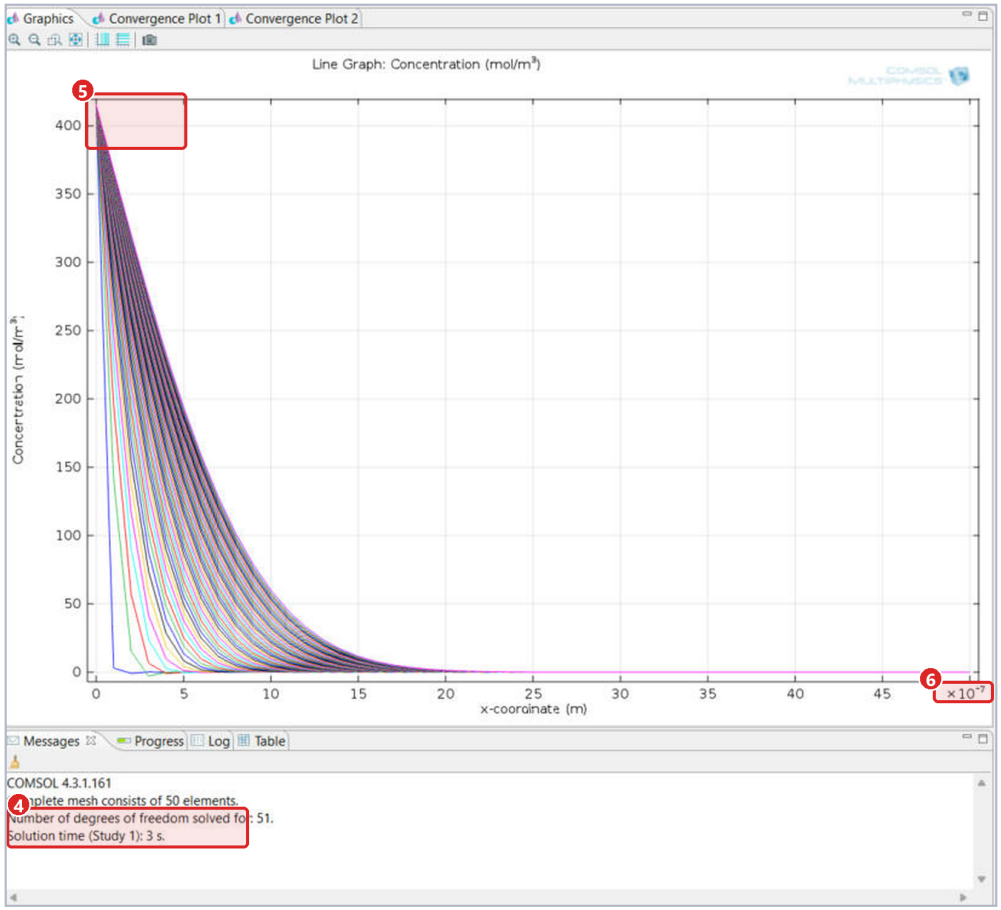
Step 11 - Set the times and run the study

We want results every minute for one hour.



1. Click Step 1: Time Dependent under Study 1.
2. Times: range(0,60,3600) that is from 0 to 3600 s in steps of 60 s (the unit s is shown on the right). These are the output times; COMSOL chooses its own internal time steps.
3. Click Compute (the green = in the toolbar, or press F8).

Step 11 - Set the times and run the study



4 Messages: Number of degrees of freedom solved for: 51. Solution time (Study 1): 3 s.

5 The default plot shows the concentration profile for every output time. The profile near the surface rises with time.

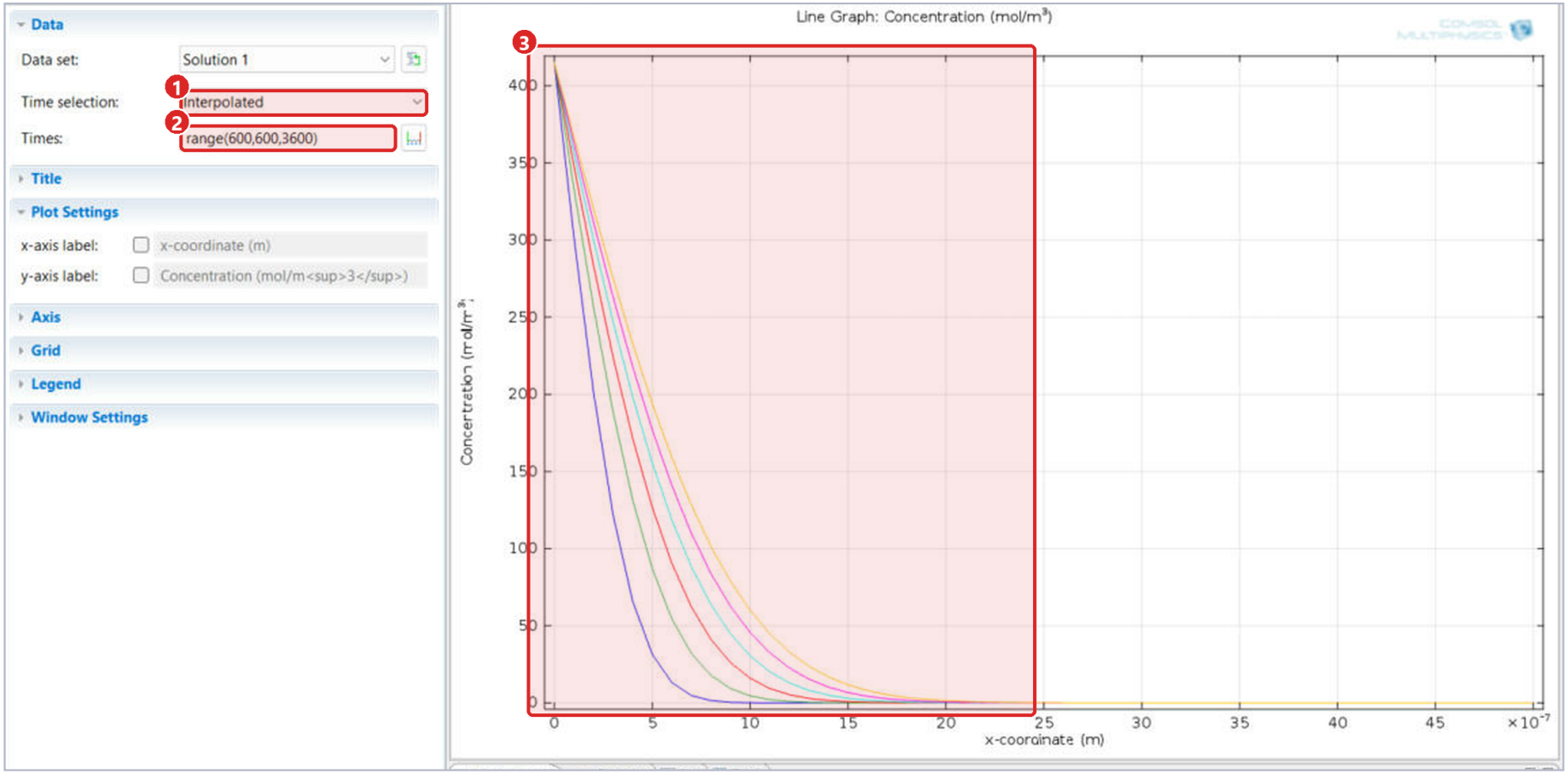
6 The x-axis carries a factor x10^-7 (metres): 50 means 5 um.

Why?

The implicit time stepper that COMSOL uses is unconditionally stable. The explicit FTCS scheme of the finite-difference task is stable only for dt <= dz^2 / (2 D) = 76.9 s (dz = 0.1 um), see the results page.

Step 12 - Plot the profiles at selected times

The default plot has 61 curves. We keep six: every 10 minutes.

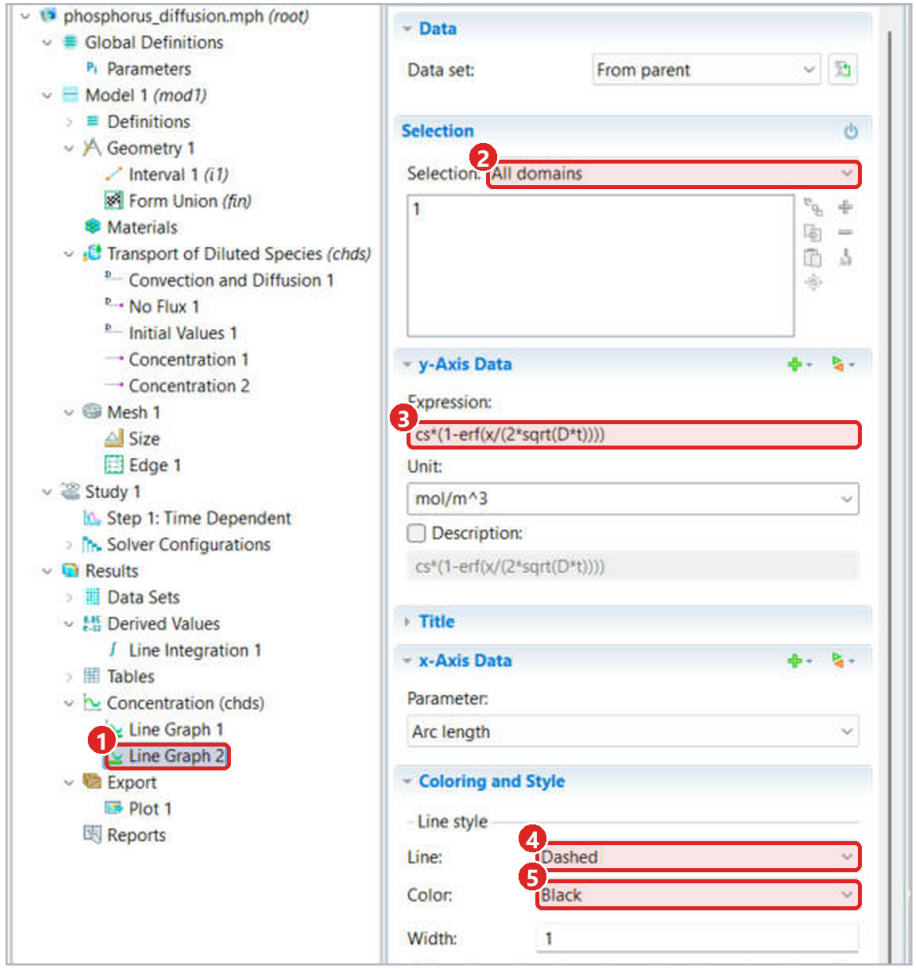


1. Click the plot group Concentration (chds). Set Time selection to Interpolated.
2. Times: range(600,600,3600) = 600, 1200, ..., 3600 s. Click Plot (F8) to update.
3. The six curves are the profiles after 10, 20, 30, 40, 50 and 60 minutes. The phosphorus penetrates deeper with time and the profile at the surface stays at 415 mol/m^3 (1 on a normalised scale).

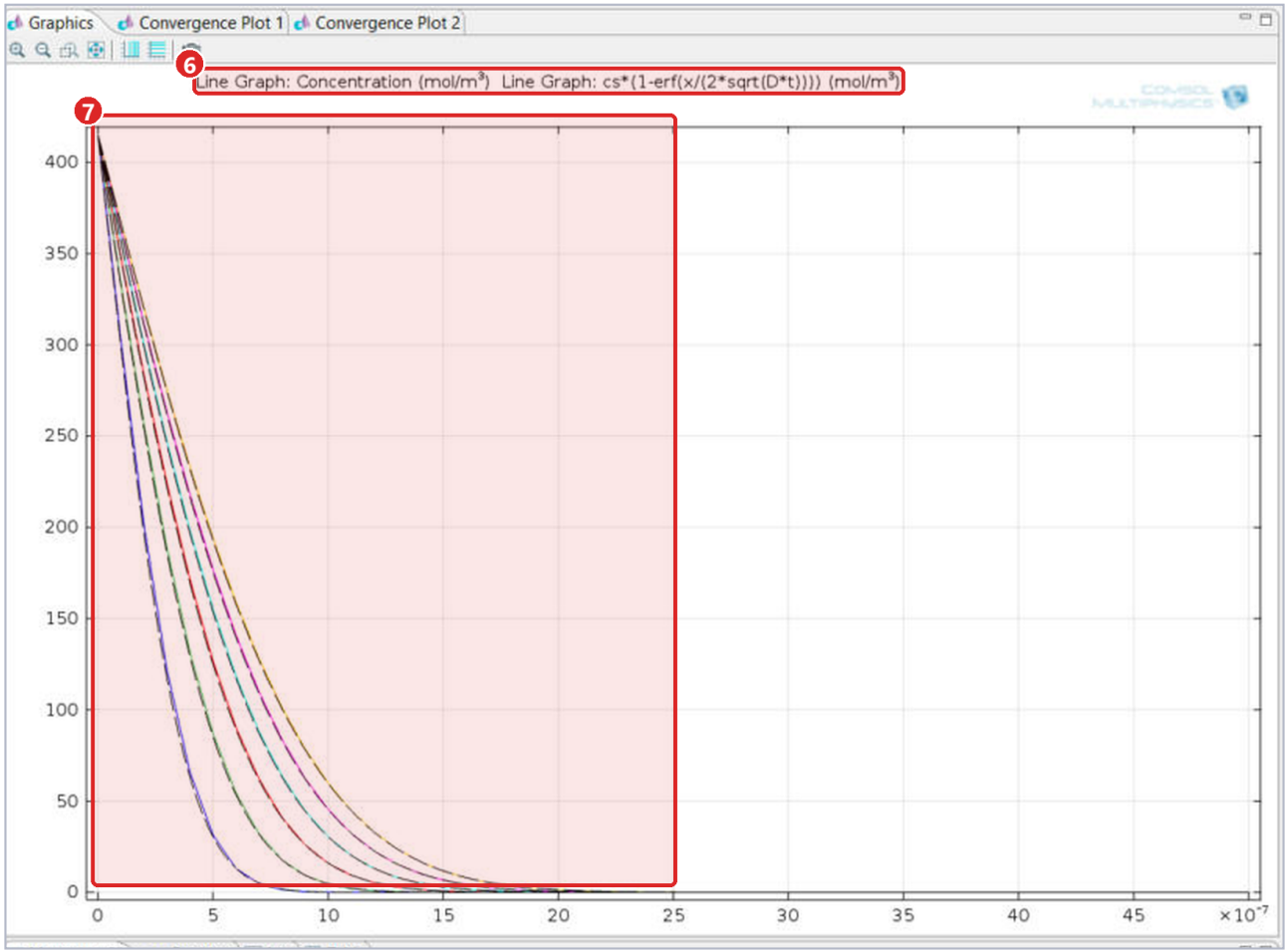
Why?
The depth reached grows like sqrt(t): quadrupling the time doubles the penetration. Compare with the exact curves in the next step.

Step 13 - Overlay the exact (error-function) solution

The exact solution of the problem is c = c_As (1 - erf(z / (2 sqrt(D t)))). We plot it as dashed lines on top of the COMSOL curves.



1. Right-click the plot group Concentration (chds) and add a second Line Graph (Line Graph 2).
2. Selection: All domains.
3. Expression: cs*(1-erf(x/(2*sqrt(D*t)))) . In the plot, x is the depth z and t is the time of each curve. (COMSOL 4.3a has no erfc function, so we use 1 - erf.)
4. Under Coloring and Style set Line = Dashed (the default is Solid).
5. Set Color = Black (the default is Cycle), then click the plot group and press F8 to plot again.



6 The plot title now lists both graphs.

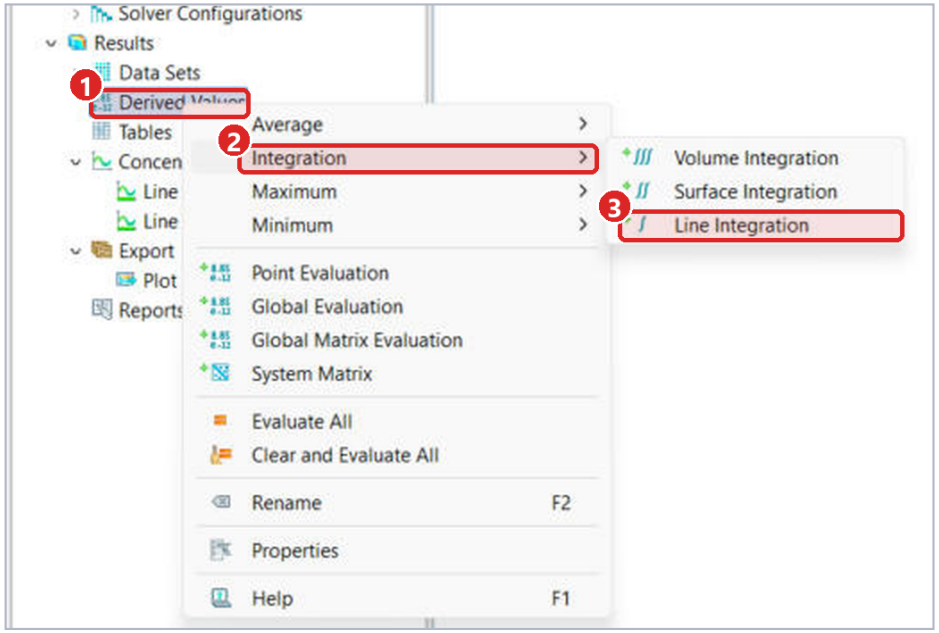
7 The black dashed exact curves lie on the COMSOL curves: the numerical solution is correct.

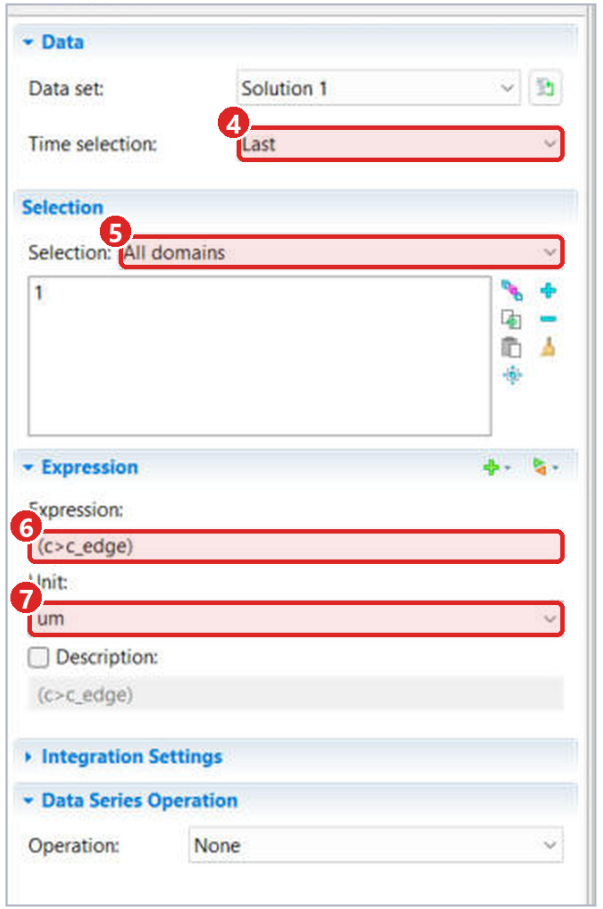
Why?

A model should always be checked against a known solution when one exists. Here the largest difference between COMSOL and the exact curve is 0.4 % of the surface value at t = 1 h; it is larger (1.9 %) at 10 min because the first minutes have a very steep profile that the default time-stepping tolerance (0.01) resolves less precisely.

Step 14 - Find the film depth after 1 h

The film depth is the depth where the phosphorus concentration has dropped to 1 % of the surface value (c_edge). We compute it with a line integral of a logical expression.





1. Right-click Derived Values.
2. Choose Integration, then Line Integration.
3. A Line Integration node is created.
4. Time selection: Last (t = 3600 s).
5. Selection: All domains.
6. Expression: (c>c_edge) which is 1 where the concentration is above the film edge and 0 elsewhere, so the integral is the length of the region where c > 1 %.
7. Unit: um. Click Evaluate (the = button at the top).

Step 14 - Find the film depth after 1 h



8 Result: 1.77222 um at t = 3600 s.

Why?
Result: film depth about 1.77 um after 1 h (exact value 1.762 um). The small difference (0.6 %) comes from the logical expression being integrated on a 0.1 um grid. Reading the crossing from the exported profile by linear interpolation gives 1.7624 um, which is within 0.02 % of the exact value (see the results page).

A. Film depth after 1 h (1 % of the surface value)

Method	Depth (um)	Error
Exact (erf solution)	1.7621	-
COMSOL, line integral (Step 14)	1.7722	+0.6 %
COMSOL, exported profile	1.7624	+0.015 %
FTCS, dz 0.1 um, dt 50 s	1.7643	+0.12 %

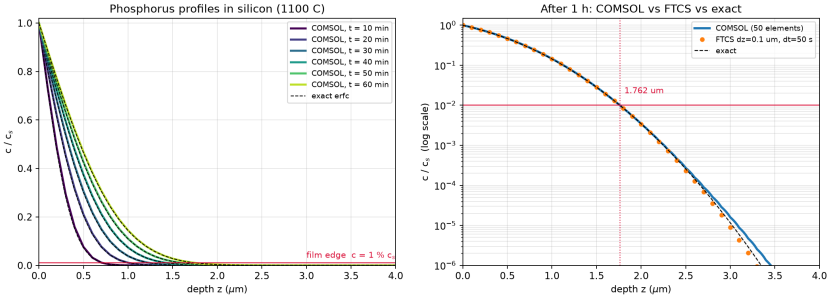
B. COMSOL vs exact profile: max |c - c_exact| / c_s (%)

Time	10 min	20 min	30 min	40 min	50 min	60 min
0.1 um mesh	1.89	1.03	0.74	0.59	0.47	0.40

C. FTCS grid refinement (Fo = D dt / dz^2 = 0.4)

dz (um)	dt (s)	steps	depth (um)	error (% cs)
0.200	240.00	15	1.7580	0.8435
0.100	61.02	59	1.7619	0.2152
0.050	15.32	235	1.7618	0.0545
0.025	3.84	937	1.7621	0.0137

Halving dz (Fourier number fixed) reduces the profile error by about 4: second order in space. COMSOL (0.4 % of c_s at 1 h) and the exact curve agree within the plot resolution.



D. Stability of the explicit (FTCS) scheme

c_i(n+1) = c_i(n) + Fo (c_(i+1) - 2 c_i + c_(i-1)), Fo = D dt / dz^2. Stability requires Fo <= 1/2, i.e. dt <= dz^2/(2 D) = 76.9 s for dz = 0.1 um.

dt = 50 s (Fo = 0.325): stable, depth 1.7643 um.
dt = 100 s (Fo = 0.65): the maximum concentration after 1 h is 3.8e7 mol/m^3, about 10^5 times the surface value of 415 mol/m^3: the solution has blown up.
dt = 80 s (Fo = 0.52): the growth is slower and needs more steps to show.
COMSOL's implicit time stepper has no such limit.

Troubleshooting and tips

The right-click menu does not appear.

Left-click the node once to select it, then right-click again.

A selection list stays empty after clicking in the Graphics window.

Hover the cursor over the point until it turns red, click it, then press the + button (Add to Selection). If it does not turn red, move the cursor slightly and click again.

The parameter value is red or looks wrong.

A red expression means COMSOL does not know a symbol. In COMSOL 4.3a use 6.02214e23[1/mol] instead of N_A, and 1-erf(...) instead of erfc(...). Units must be inside square brackets.

The depth results look wrong by a factor 10^6.

COMSOL works in metres. Set the unit of the derived value to um (type um in the Unit field) or multiply by 1e6.

The solution oscillates or goes negative at early times.

This is a numerical effect of the very steep profile in the first seconds. Use a finer mesh near the surface (smaller element size) or reduce the relative tolerance of the time-dependent solver.

COMSOL asks for a license.

Your computer must be on the university network or VPN so it can reach the license server.

Save your work.

File > Save As, choose a folder you can write to. Compare with the supplied file phosphorus_diffusion.mph.